Jackpot user guide

Jackpot is **the casino floor for Resolume Arena and Avenue**, as two FFGL plugins: **SW Jackpot**, a source, and **SW Jackpot Over**, the same over your clip. A one-armed bandit, a roulette wheel, the money wheel, a craps table and the lottery drum — and chips and coins showering down on a win. Press **Play**: the game plays out with real physics and comes to rest on a random result, or on the one you set, at exactly the Spin Time you set, or on the next beat or bar.



Three sevens on the line, the banner, the meters counting the win in, and the shower a jackpot sets off. Rendered by the plugin's offline harness, not captured from Resolume.

> **Before you rely on this:** released at **v0.1.1**, and honestly early. It has been loaded into Resolume Arena 7.27.1 on a Mac and played there — a slot jackpot and a roulette spin, each landing as asked, with the shower. On Windows, both plugins load, register and render in Resolume Arena 7.27.1, with every control as declared, on a machine with software rendering and no GPU; no game was played there. Everything else is measured by a harness that drives the real plugin classes in a headless GL context. Every number of both roulette wheels, every value of the money wheel and every craps total was played, and each came to rest as asked; the harness then reads the result **out of the rendered picture** for the slot machine, the roulette wheel and the money wheel. Every game stops at exactly Spin Time. Twenty-four deliberately wrong models are all detected, twelve one-character changes to the shipped code are all caught, and all 46 controls across the two plugins measurably do something. v0.1.1's shower fix was run in Resolume Arena 7.27.1 on Windows (software rendering, a build of the 0.1.1 code made with MSVC on the lab machine, 2026-10-12): with Shower Only on Always, chips showered, then coins, the chips were gone about 14 s after the switch, the shower kept falling, and the floor was clear 15 s after Shower Off. **Play**'s place at the top of the panel is measured by the harness only. Try it on a spare layer before you put it in a show.
>
> This codebase was created with AI assistance, directed and reviewed by a human author.

Installing

Every download carries **both plugins**: the source and the effect. On macOS they are `Jackpot.bundle` and `Jackpot Over.bundle`; on Windows, `Jackpot.dll` and `Jackpot Over.dll`. Put both into Resolume's effects folder and restart Resolume:

```
macOS    ~/Documents/Resolume Arena/Extra Effects/
Windows  %USERPROFILE%\Documents\Resolume Arena\Extra Effects\
```

Avenue uses the same layout under its own folder name. **SW Jackpot** then appears among the sources and **SW Jackpot Over** among the effects.

The macOS download is a universal build (Apple silicon and Intel), as a `.dmg` or a `.zip`. It is **Developer ID-signed and notarised**, so the bundles simply load. The Windows download is an x64 installer or a `.zip`. It is not code-signed, so the installer trips SmartScreen once: **More info** → **Run anyway**.

Two plugins

- **SW Jackpot** (a source) draws the game on its own. With **Backdrop** on *Dark* — the default — it stands in a dark room with the casino's lights out of focus behind it; *Felt* puts it on a green table; *None* draws nothing behind the game, so it goes over whatever is underneath.

- **SW Jackpot Over** (an effect) draws it over the clip it is on. Its **Symbols** list has one more entry, **Clip**, which puts the clip itself on the slot machine's reels where the seven would be.

The effect declares every one of the source's controls and adds **Mix** at the end, after About. A look built on one carries across to the other.

How a result is chosen

Every casino game is decided before it is shown — and every one has something the physics cannot tell apart. A roulette rotor's thirty-seven pockets are identical; the money wheel's fifty-four pegs are identical; a die looks the same from every face; lottery balls are all alike. Jackpot uses that. When you press Play it plays the game honestly, ahead of time — the ball on the track, the wheel against its clapper, the dice off the back wall, the balls in the air — until it comes to rest. It reads where it came to rest, and then turns **the paint**: the numbered ring, the painted segments, the die's faces, which number is printed on which ball, so that what you see there is the result chosen.

Every bounce, every tick of the clapper and every ball you watch is the physics' own; only which number is painted where has moved, and you never see it move: on the roulette wheel the croupier's spin of the rotor before the ball is released takes the ring there, on the money wheel it is the pull before it is let go, and the lottery's drum is empty between draws, so the new balls drop in already numbered.

The slot machine needs none of that: since the 1980s a slot's result has been decided by its random number generator the moment the handle goes down, and the reels have been a display. The plugin does exactly that and drives the reels to the stops.

So a chosen result looks exactly as random as a Random one, and a Random result is drawn by the plugin, weighted as the real game is, not left to how the physics happened to go.

Start here

1. Put **SW Jackpot** on a layer. You see a red fruit machine in a dark room.
2. Press **Play** (map it to a key, a MIDI note or a Stream Deck button). The handle goes down, the reels spin and stop one after another; the last stops at Spin Time, 4 seconds later. A win lights the banner, counts the credits in and showers chips.
3. Choose a **Game**: *Slots*, *Roulette*, *Money Wheel*, *Craps*, *Lottery*, or *Shower Only*.
4. For a chosen outcome, set **Result** — *Win*, *Jackpot*, *Near Miss*, *Lose* or *Fixed* — and, where it applies, **Fixed Number**. Press Play: the game lands on it.
5. Set **Land On** to *Beat* or *Bar* to have every play come to rest on the beat.

On roulette, the money wheel, craps and the lottery the play starts a frame or two after the press: the game is worked out on another thread first (about 1 ms for craps, 13 for roulette, 50 for the lottery and the money wheel).

Game, Play and the Game group

Game and **Play** stand at the top of the panel, outside every group, so Play is there to press with all the groups closed. The **Game** group under them holds the rest, from Spin Time to Interval.

- **Game** — *Slots*, *Roulette*, *Money Wheel*, *Craps*, *Lottery*, *Shower Only*. Changing it lays the new game out at once, at rest.
- **Play** — the button. A press while a game is still playing starts another from where it is. On *Shower Only*, Play is a burst of chips.
- **Spin Time** — 1 to 20 seconds from Play to rest, default 4. See *Spin Time* below.
- **Land On** — *Time* (exactly Spin Time), *Beat* or *Bar*: the first beat or bar of Resolume's tempo at least Spin Time away.
- **Result** — *Random*, *Win*, *Jackpot*, *Near Miss*, *Lose*, *Fixed*. Each game reads them against the bet, which is Fixed Number:

	Slots	Roulette	Money wheel	Craps	Lottery
Random	weighted as the strip	each pocket alike	each segment alike	each die alike	each ball alike
Win	any paying line but the jackpot	the bet's number	the bet's value	a winning roll	the bet drawn
Jackpot	sevens across	the bet's number	joker or logo	11 on the come-out; the point made (the hard way on 4, 6, 8, 10)	the bet drawn last
Near Miss	two sevens, the third beside the line	a pocket beside the bet	beside the joker or logo	6 or 8 on the come-out; one off the point	the number beside the bet
Lose	nothing paid, no near miss	anything but the bet	anything but the bet or the top	craps, or seven-out	the bet not drawn
Fixed	Fixed Number is the symbol	the number (37 is 00)	the value	the total	the bet drawn first

- **Fixed Number** — 0 to 99: the bet. On the slot machine it names a symbol: 0 blank, 1 cherry, 2 lemon, 3 orange, 4 plum, 5 bell, 6 melon, 7 bar, 8 double bar, 9 triple bar, 10 seven, 11 diamond. On the money wheel a value it does not have goes to the nearest one (1, 2, 5, 10, 20; 40 is the joker, 45 the logo).

- **Seed** — 0 to 9999. The same seed replays the same sequence of plays.
- **Auto Play** and **Interval** — plays by itself, every 2 to 120 seconds (default 10). Switching Auto Play on plays at once.

The slot machine



Every game at rest on its result, with the result board above it.

A three- or five-reel fruit machine with a handle, a marquee of chasing lamps, the credit and win meters and three buttons. Each reel carries 22 stops: eleven symbols with a blank between each pair, a seven at stop 0.

- **Reels** — *3* or *5*.
- **Symbols** — *Fruit* (cherries, lemon, orange, plum, bell, melon, bar, seven), *Sevens and Bars* (single, double and triple bars, cherry, diamond, seven), and on the effect *Clip* (the fruit, with the clip where the seven would be).
- **Strip** — *Virtual* or *Physical*; see below.
- **Blur** — the camera's shutter, from a still frame (0) to the whole frame time (1, a 360° shutter). At speed the symbols smear into streaks and sharpen as the reel brakes.
- **Reel Bounce** — how lively the detent is when it catches a reel: from dead (0) to a few springy bounces (1).

It pays from the left: three or more of a symbol — seven 200, diamond 100, triple bar 60, double bar 40, bar 20, bell 18, melon 15, plum 14, orange 10, cherry 10, lemon 8; four in a row pays four times that and five twenty times — any three bars mixed 10, a cherry on the first reel 2 and two cherries 5. A pull costs a credit; the meter starts at 100.

The near miss is real. On the *Virtual* strip a Random pull draws each reel's stop from 64 virtual stops mapped onto the 22 physical ones unevenly: the seven gets one, and each blank beside it gets six. So the seven goes sailing past just above or below the line twelve times for every time it

lands on it — twice on a fair strip — because that is how slot machines have been built since Inge Telnaes's patent in 1984. *Physical* makes every stop equally likely.

The roulette wheel

A wooden bowl, the ball track and its lip, the stator with eight diamond deflectors, the rotor with its numbered ring, its pockets and the frets between them, and the turret. The ball runs round the track against the lip until it is too slow to stay up, spirals down the stator, can strike a diamond, rattles across the frets of the rotor turning the other way, and settles.

- **Wheel** — *European* (37 pockets, a single zero) or *American* (38, with 00), in their real order.
- **Rotor Speed** — how fast the croupier spins the rotor, up to about 38 turns a minute.
- **Deflectors** — the eight diamonds on the stator, on or off.

The croupier spins the rotor up before releasing the ball, and the ring turns on between plays. The result board shows the number and its colour once the ball is at rest; the winning pocket glows.

The money wheel

The Big Six: a big wheel of 54 segments — 24 ones, 15 twos, 7 fives, 4 tens, 2 twenties, a joker and a logo — a brass peg between each pair, and a leather clapper at the top that the pegs push past. Each peg bends the leather and takes energy from the wheel, so it ticks slower and slower and may rock back off the last peg.

- **Clapper** — how stiff the leather is. Stiffer takes more from each peg and stops the wheel sooner, so the operator pulls it harder.

The craps table

The end of the table by the back wall: the layout (the point boxes 4, 5, SIX, 8, NINE, 10; COME; FIELD; DON'T PASS BAR; PASS LINE), the rubber pyramids on the back wall and the puck. The stick sweeps the last roll's dice away, the shooter throws, the dice hit the back wall and come to rest on the layout.

- **Pyramids** — the back wall's rubber pyramids, on or off.
- **Puck** — the ON/OFF puck, shown or not.

The table keeps its point from roll to roll: a come-out 7 or 11 wins, 2, 3 or 12 is craps, anything else is the point and the puck goes ON; then the point again wins and a 7 is a seven-out. Result reads against the point the table is on.

The lottery drum

A glass drum of numbered balls in the UK lotto's colours, a jet of air up the middle that carries them to the top where they fall back down the glass, a tube at the top and a rack. The old balls drain out of the bottom, the new ones drop in, the air comes on, and the tube takes one ball at a time up to the rack.

- **Balls** — 10 to 75 in the drum (default 49).
- **Draw** — 1 to 7 balls drawn (default 6).
- **Air** — how hard the blower blows.

It takes about two seconds just to load 49 balls, so a draw shorter than about six seconds plays fast.

The shower

Chips and coins, each one a real spinning disc: they tumble in the air, bounce, roll on their edges and, as they lean over, whirr faster and faster before they slap flat — Euler's disk — and land in a pile along the foot of the frame.

- **Shower** — *Off*, *On Win* (the default), *On Jackpot*, or *Always*.
- **Shower Now** — a burst, whatever the game is doing.
- **Pattern** — *Rain* across the frame, a *Fountain* from the bottom, a *Burst* from the middle, or a *Pour* from one place.
- **Pieces** — *Chips*, *Coins* or *Mixed*.
- **Amount** — how many fall each second, up to about 400.
- **Piece Size** — a chip's width as a share of the frame's height, 2% to 20%.
- **Pile** — on, the pieces settle into a pile; off, they fall out of the frame, and the pile there fades away in half a second.

The pile clears itself. Every piece fades twelve seconds after it was thrown — about ten seconds lying in the pile — oldest first, so a win's pile is gone about fifteen seconds after the shower began, and a run of wins keeps one pile that renews itself. A heavy or constant shower fills the plugin's 900 pieces quickly; then the oldest go early, in a fifth of a second, so the shower never stops for want of room. A piece that rolls off the table, at the sides, the front or the back, is gone. Changing **Pieces** does not sweep the table: the old kind fades out with its age, within about twelve seconds. The twelve seconds are fixed, not a control.

Shower Only as the Game is the shower on its own: Play is a burst.

The Look group

- **Backdrop** — *None* (nothing behind the game: transparent on the source, the clip on the effect), *Felt* (a green table) or *Dark* (a dark room with the casino's lights out of focus).
- **Felt Colour** — the table's cloth, on the felt and on the craps layout.
- **Accent Colour** — the slot machine's cabinet and the money wheel's rim.
- **Lights** — the lamps: the marquee's chase, the money wheel's rim, the glow on a winner.
- **Display** — the slot machine's banner and the other games' result board, on or off.
- **Font**, **Font File**, **Font Name** — the type on the meters, the banner, the layout and the numbers: *Built-in* (the same on every machine), any installed font, or a font file. **Fonts travel by name:** the list's position means a different font on another computer, so the family is kept in Font Name, and when a composition restores both, the name wins.
- **Light Angle** — where the key light comes from, round the scene (roulette, craps, the lottery and the shower).
- **Tilt** — how high the camera looks from, 20 degrees to straight down (roulette, craps and the lottery; the lottery is always seen 35 degrees lower, from in front).
- **Zoom** — closer or wider.

SW Jackpot Over

Everything above, over the clip. **Mix** fades between the clip alone (0) and the clip with the game (1). Where the game draws nothing the clip comes through untouched, its alpha too.

Symbols: Clip puts the clip on the slot machine's reels in place of the seven, so a jackpot is three of your clip.

Spin Time

Every game comes to rest at exactly Spin Time (or on the beat or bar it was sized for). Each game first aims its own natural length at it — the ball's speed, the pull on the money wheel — and then plays what it simulated a little faster or slower so it stops on time, easing in slow motion.

- **Slots** is always at real speed: the reels are driven to stop at Spin Time.
- **Roulette** and **the money wheel** are thrown to take about Spin Time on their own.
- **Craps** never throws harder to fill time. Dice thrown down a table settle in under a second however hard they are thrown, so a long Spin Time is the shooter holding the dice a moment after the sweep and at most a gentle slow motion.
- **The lottery** needs time to load and mix; a short draw plays fast.

Time comes from the host

The games move on the host's clock, so a paused composition freezes a play. A step backwards in the host's clock, or a jump forward of more than a quarter of a second (a clip trigger or a scrub), passes no time: the play carries on from where it was. For the first few frames after it loads, the plugin runs on its own steady clock while it works out whether the host counts time in seconds or milliseconds.

Performance

Measured by the offline harness on an Apple M4 Max, mid-play, the median frame, on a machine shared with other builds:

	720p	1080p	4K
slots	0.5 ms	0.8 ms	1.7–1.8 ms
roulette	2.0–2.2 ms	4.1–4.2 ms	5.3–5.4 ms
money wheel	0.3 ms	0.7 ms	1.4 ms
craps	1.2–1.3 ms	2.2–2.3 ms	2.5–2.6 ms
lottery, 49 balls	0.6–0.7 ms	1.2–1.5 ms	3.4 ms
a constant shower	1.0 ms	1.2 ms	2.3 ms

Up to 1080p every pixel of roulette, craps and the lottery takes four rays; above it only the pixels on an edge do. Nothing was timed inside Resolume.

If it looks wrong

Nothing happens when I press Play. On roulette, the money wheel, craps and the lottery the play starts a frame or two after the press, and comes to rest at Spin Time; at 20 seconds it is a long spin. A paused composition freezes it.

I asked for Jackpot and got something else. Check the Result row really reached the plugin: the log records every play's result (below). Then check Fixed Number: on roulette Jackpot and Win are the bet's number, and on the money wheel a value the wheel does not have goes to the nearest one.

The lottery plays very fast. Spin Time is shorter than loading and mixing the balls takes. Give it six seconds or more.

The dice are in slow motion. Spin Time is longer than a throw; the plugin plays the throw up to a little slower and holds the dice before it, but no longer than that.

My font changed on another computer. The font is not installed there; the text uses the built-in face, and Font Name still holds the name.

The effect does nothing at all. A shader that will not compile looks exactly like that. The real message is in the log:

```
macOS    ~/Library/Logs/jackpot/jackpot.YYYY-MM-DD.log
Windows  %LOCALAPPDATA%\jackpot\logs\jackpot.YYYY-MM-DD.log
```

It records the GL vendor, renderer and version at load, a shader that failed, the font in use (and a named font that is not installed), and every play: the game, its result, and how much it was slowed or sped up.

What is verified, and what is assumed

In Resolume Arena 7.27.1 on a Mac (v0.1.0): both plugins load and register; SW Jackpot in a clip shows every control; a slot Jackpot and a roulette spin landed as asked, with the shower and the result board.

In Resolume Arena 7.27.1 on Windows (software rendering, no GPU): both plugins load and register, every control is as declared, both render with a font loaded from file, and Resolume's log stays clean. Every control that can change the picture without a play was seen doing so (25 on SW Jackpot, 27 on SW Jackpot Over). No play was pressed there, so the games themselves have not run on Windows.

Measured, on an M4 Max under macOS, by the offline harness driving the real plugin classes: every reel at rest exactly on its stop, every Result honoured and the pay table as written above; the near miss twelve times a jackpot on the virtual reel and twice on the physical one; the reels' blur the strip integrated over the shutter; the payline read out of the picture; every number of both roulette wheels at rest under the ball, and read out of the picture; turning the rotor by whole pockets leaving the ball's path identical to the bit, and the ball leaving the track at the speed the physics of a banked track says; every money-wheel value under the clapper, and the clapper taking the energy; every craps total on the dice as drawn, and the pass line; the lottery's rack showing the numbers drawn; the chips' fall, tumble, bounce, pile and Euler's whirr against the laws they should obey; the pile clearing within thirty seconds of the last shower, no piece left off the table, a constant shower at full Amount still throwing everything asked of it after a minute, and the table flat again once the pile has gone or Pile is switched off and on; Game and Play outside every group; Random fair by chi-square; every game stopping at Spin Time and on the beat; the clip untouched by the effect where it must be. Every check also runs against a deliberately wrong model and must fail it, and a sweep fails if any control does nothing.

Assumed, or not verified:

- **Two plays in one Resolume, on one Mac, and none on Windows.** The Over effect, the money wheel, craps and the lottery, Land On against Resolume's tempo, and a saved composition's restore have not been tried in a host, nor any play on Windows or on a GPU there.
- **Play's place at the top of the panel (v0.1.1)** is measured only by the harness. That Resolume draws Game and Play outside the groups follows from how it groups parameters; it has not been looked at. The pile's clearing was run in Resolume Arena 7.27.1 on Windows (software rendering, a build of the 0.1.1 code made with MSVC on the lab machine, 2026-10-12), with Shower Only, not after a game.
- **The physics is its own**, checked against the laws it should obey, not against film of real casino equipment. Every friction, bounce and stiffness is chosen to look right.
- **The roulette frets are flush with the pockets' rims**, and the ball is a point carrying its radius.
- **A piece that comes to rest within 16 degrees of flat is laid flat** (by up to about eleven, in practice); steeper, leaning on a step of the pile, it stays as it lies. The pile is a height field, not a stack of bodies: when a piece fades, those lying on it drop onto what is under it.
- **The same seed gives the same results everywhere**, because they come from the plugin's own random numbers or your bet. The motion itself may differ between a Mac and a Windows machine.
- **The inspector shows slider positions**, not seconds or degrees. This guide gives the ranges.
- **No OpenFX version, no presets.**

About

The last group in the source's panel, **About**, carries the plugin's name, version, licence and maker, and buttons that open this guide, the project page, the source on GitHub and the support page in your browser. In the effect, Mix follows it.

Links

- Project page: stoatworks-labs.com/software/jackpot
- Source: github.com/stoatworks-labs/jackpot

Reporting something

github.com/stoatworks-labs/jackpot/issues. A screenshot, the Game and Result, any controls you moved, whether it was the source or the effect, and the composition's resolution and frame rate are usually enough.